NUMERICAL METHODS FOR PDES
Lecture 6

FD in 2D (recap)

$$\begin{cases} -\Delta u = f & \text{in } (0,1)^2 = \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$$

Central FD method

$$\begin{cases} -\dfrac{U_{i+1,j} + U_{i,j+1} - 4U_{ij} + U_{i-1,j} + U_{i,j-1}}{h^2} = f_{ij} \quad (= f_h U_{ij}) \\ U_{ij} = 0 \quad \text{otherwise} \end{cases} \qquad 1 \le i,j \le N-1$$

Lemma (consistency): If $u \in C^4(\Omega) \cap C(\bar{\Omega})$, then

$$|T_{ij}(\underline{x})| \le \frac{h^2}{12}\left(M_{xxxx} + M_{yyyy}\right)$$

$$\underset{\|}{L(\underline{x})} - L_h(\underline{x}) = f(\underline{x}) - L_h(\underline{x})$$

$\underline{x} = (x, y)$

Call $T_{ij} = T(x_i, y_j)$

Lemma (Stability): For any grid function $V = \{V_{ij}\}_{ij}$, it holds

$$\|V\|_{l_\infty(\Omega)} \leq \|V\|_{l_\infty(\partial\Omega)} + \frac{1}{8}\|L_h V\|_{l_\infty(\Omega)}$$

Proof: again by comparison function,

$$\omega(x) = x + y - x^2 - y^2$$

Theorem (convergence): If $u \in C^4(\Omega) \cap C^0(\bar{\Omega})$, then the

5-point centered FD solution U for any $h>0$, satisfies

$$|u(x_i,y_j)-U_{ij}| \le \frac{h^2}{96}\left(M_{xxxx}+M_{yyyy}\right)$$

proof: let $e_{ij}=u(x_i,y_j)-U_{ij}$

and $E=\{e_{ij}\}_{ij}$

$$|e_{ij}| \le \underbrace{1}_{\text{stability}} \|E\|_{\ell_\infty(\partial\Omega)} + \frac{1}{8}\|\delta_h E\|_{\ell_\infty}$$

$\|E\|_{\ell_\infty(\partial\Omega)} = 0$ using error is zero at boundary

$$\delta_h e_{ij} = \delta_h\left(u(x_i,y_j)-U_{ij}\right)$$

$$= \delta_h\left(u(x_i,y_j)\right) - f_{ij} = T_{ij}$$

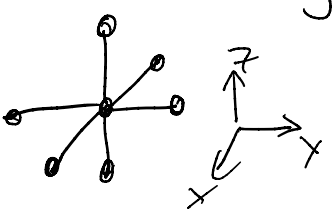
by FD scheme

$$\Rightarrow |e_{ij}| \le \frac{1}{8}\frac{h^2}{12}\left(M_{xxxx} + M_{yyyy}\right) \quad \square$$

Note: extension

1) 3D case trivial $\Rightarrow$ 7-points scheme

$$-(u_{xx}+u_{yy}+u_{zz}) = f$$

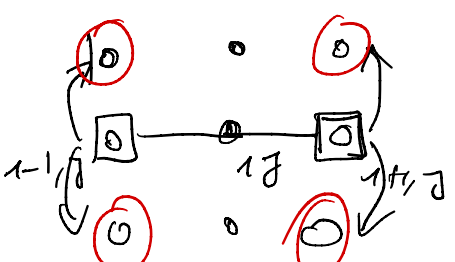


2) For, say

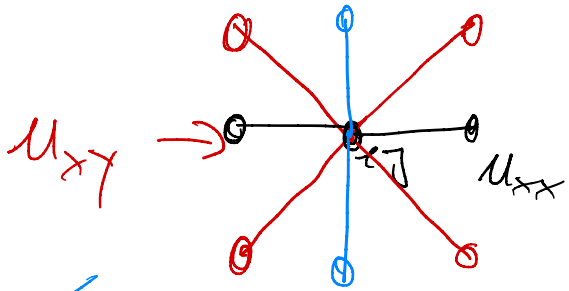
$$a u_{xx} + c u_{yy} + 2b u_{xy} = f$$

$$+u_x + u_y + u$$

$$u_x \sim \frac{U_{i+1,j} - U_{i-1,j}}{2h}$$



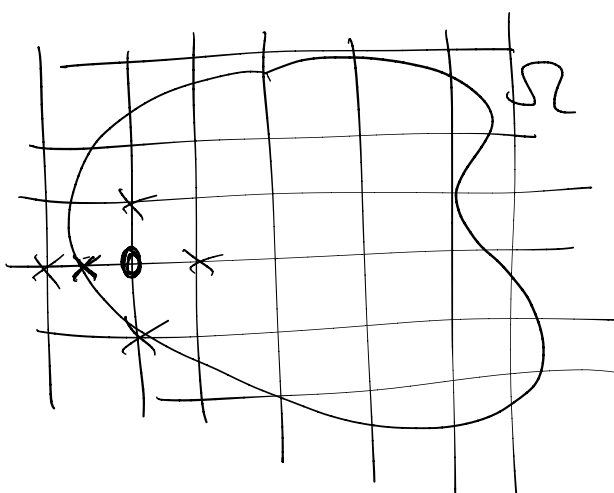
$$u_{xy} \quad \frac{U_{i+1,j+1} - U_{i+1,j-1} - U_{i-1,j+1} + U_{i-1,j-1}}{4h^2}$$



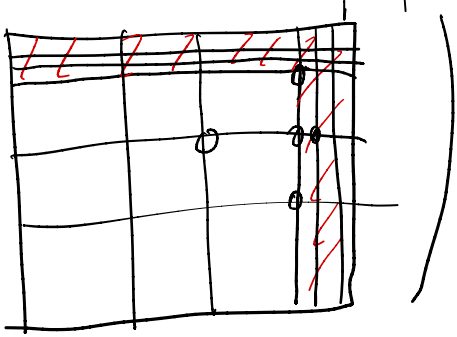
2D 9-points stencil

3) Extension to general domains?

→ use nonuniform grids



(also useful to adapt the grid to the solution)



going back to

$$u'' = f \qquad (0,1)$$

on nonuniform partition $x_i = x_{i-1} + h_i$

$x_0 = 0$ and $\sum h_i = 1$

$$\frac{2}{h_{i+1}(h_i + h_{i+1})} U_{i+1} - \frac{2}{h_i h_{i+1}} U_i + \frac{2}{h_i(h_i + h_{i+1})} U_{i-1} = -f_i$$

3-points formula

- if $h_i = h_{i+1} = h \Rightarrow$ reduces to 2nd CD formula.
- in general,

$$|T_i| \le \underbrace{\frac{2}{3} h_{max}}_{\text{1st order in general if}} |u'''(x_i)| + \boxed{\frac{3}{2}\frac{h_{max}}{h_{min}}} M_{xxxx}$$

$\boxed{} \to 0$

9 Functional analytic preliminaries

Throughout, $\Omega \subset \mathbb{R}^d$ is open. It will be assumed bounded when an embedding or estimate requires $|\Omega| < \infty$, and Lipschitz when traces on $\partial\Omega$ are used. Functions that agree almost everywhere are identified.

9.1 The spaces $L^p(\Omega)$

For $1 \le p < \infty$, define

$$L^p(\Omega) = \left\{ v : \Omega \to \mathbb{R} \text{ measurable} : \int_\Omega |v|^p \, \mathrm{d}x < \infty \right\},$$

with norm

$$\|v\|_{L^p(\Omega)} = \left(\int_\Omega |v|^p \, \mathrm{d}x \right)^{1/p}.$$

The limiting space is

$$L^\infty(\Omega) = \left\{ v : \Omega \to \mathbb{R} \text{ measurable} : \operatorname*{ess\,sup}_{x \in \Omega} |v(x)| < \infty \right\},$$

with

$$\|v\|_{L^\infty(\Omega)} = \operatorname*{ess\,sup}_{x \in \Omega} |v(x)| .$$

The essential supremum, rather than the pointwise supremum, makes the norm independent of changes on sets of measure zero.

Let p' be the conjugate exponent, defined by

$$\frac{1}{p} + \frac{1}{p'} = 1,$$

with the usual conventions $1' = \infty$ and $\infty' = 1$.

Theorem 9.1 (Hölder inequality). *If $v \in L^p(\Omega)$ and $w \in L^{p'}(\Omega)$, then*

$$\left| \int_\Omega vw \, \mathrm{d}x \right| \le \|v\|_{L^p(\Omega)} \, \|w\|_{L^{p'}(\Omega)} .$$

When $|\Omega| < \infty$, Hölder's inequality also allows to compares different Lebesgue spaces. If $1 \le p \le q \le \infty$, then

$$L^q(\Omega) \hookrightarrow L^p(\Omega), \qquad \|v\|_{L^p(\Omega)} \le |\Omega|^{1/p - 1/q} \, \|v\|_{L^q(\Omega)} . \tag{9.1}$$

The assumption $|\Omega| < \infty$ is essential.

9.2 The Hilbert space $L^2(\Omega)$

The space $L^2(\Omega)$ is a Hilbert space with inner product

$$(u, v)_{L^2(\Omega)} = \int_\Omega uv \, \mathrm{d}x.$$

The induced norm is the usual L^2-norm, and Hölder's inequality with $p = p' = 2$ becomes the Cauchy–Schwarz inequality

$$\left|(u, v)_{L^2(\Omega)}\right| \leq \|u\|_{L^2(\Omega)} \, \|v\|_{L^2(\Omega)} \, .$$

Unlike continuous functions, L^2-functions may have jumps and need not possess pointwise values with any intrinsic meaning.

9.3 Dual spaces and the Riesz theorem

Let V be a normed space. Its dual space

$$V' = \mathcal{L}(V; \mathbb{R})$$

consists of the bounded linear functionals $F : V \to \mathbb{R}$, equipped with

$$\|F\|_{V'} = \sup_{v \in V \setminus \{0\}} \frac{|F(v)|}{\|v\|_V}.$$

The value $F(v)$ is also written as the duality pairing $\langle F, v \rangle_{V',V}$.

Theorem 9.2 (Riesz representation). *Let H be a real Hilbert space. For every $F \in H'$, there is a unique $u_F \in H$ such that*

$$F(v) = (u_F, v)_H \qquad \forall v \in H,$$

and $\|F\|_{H'} = \|u_F\|_H$.

In particular, every functional on $L^2(\Omega)$ can be represented by a unique L^2-function:

$$F(v) = \int_\Omega fv \, \mathrm{d}x.$$

This justifies the customary identification $(L^2(\Omega))' \cong L^2(\Omega)$. For a general space V, however, elements of V' need not be ordinary functions.

$\langle F, v \rangle = (f, v)$

$\exists! f$

9.4 Test functions and distributions

The space of test functions is

$$\mathcal{D}(\Omega) = C_c^\infty(\Omega).$$

A sequence φ_n converges to φ in $\mathcal{D}(\Omega)$ if all functions have support in one compact subset of Ω and every derivative converges uniformly there. This is the standard test-function topology.

Definition 9.3 (Distribution)**.** A distribution on Ω is a continuous linear functional $T : \mathcal{D}(\Omega) \to \mathbb{R}$. The space of distributions is denoted by $\mathcal{D}'(\Omega)$, and its action is written

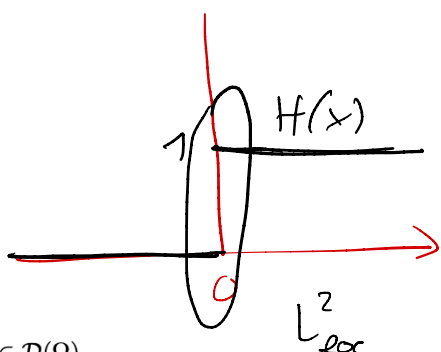
$$\langle T, \varphi \rangle = T(\varphi).$$

Every $f \in L^1_{\text{loc}}(\Omega)$ defines a regular distribution by

$$\langle f, \varphi \rangle = \int_\Omega f\varphi \, \mathrm{d}x.$$

Thus $L^p(\Omega) \subset \mathcal{D}'(\Omega)$ for every $1 \le p \le \infty$. Not every distribution is regular: for example, the Dirac mass at $x_0 \in \Omega$ is defined by

$$\langle \delta_{x_0}, \varphi \rangle = \varphi(x_0).$$



9.5 Distributional derivatives

For $T \in \mathcal{D}'(\Omega)$, define its distributional derivative by

$$\langle D_i T, \varphi \rangle = -\langle T, D_i \varphi \rangle \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

More generally, for a multi-index α,

$$\langle D^\alpha T, \varphi \rangle = (-1)^{|\alpha|} \langle T, D^\alpha \varphi \rangle . \tag{9.2}$$

Every distribution therefore has derivatives of every order, although those derivatives need not be representable by locally integrable functions.

Example 9.4 (The Heaviside function)**.** On $\mathbb{R}$, let $H(x) = 1$ for $x > 0$ and $H(x) = 0$ for $x < 0$. Then

$$\langle DH, \varphi \rangle = -\int_0^\infty 1 \cdot \varphi'(x) \, \mathrm{d}x = \varphi(0) = \langle \delta_0, \varphi \rangle .$$

Hence $\boxed{DH = \delta_0}$ in $\mathcal{D}'(\mathbb{R})$, even though H has no classical derivative at the origin.

The above examples shows that if a function belongs to L^p, its derivatives, which are always defined in the distributional sense, may *not* belong to L^p itself.

$\langle H, \varphi' \rangle$ $\quad \delta_0 \notin L^2_{loc}$

9.6 Sobolev spaces

Definition 9.5 (Sobolev space)**.** For $k \in \mathbb{N}_0$ and $1 \le p \le \infty$, define

$$W^{k,p}(\Omega) = \{v \in L^p(\Omega) : D^\alpha v \in L^p(\Omega) \text{ for all } |\alpha| \le k\},$$

where all derivatives are understood in the distributional sense.

When $p = 2$, write $H^k(\Omega) = W^{k,2}(\Omega)$. It is a Hilbert space with

$$(u, v)_{H^k(\Omega)} = \sum_{|\alpha| \le k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)}.$$

In particular, $k=1$

$$H^1(\Omega) = \left\{v \in L^2(\Omega) : \nabla v \in L^2(\Omega)^d\right\}, \qquad \|v\|^2_{H^1(\Omega)} = \|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)}.$$

9.7 Traces, zero boundary values, and Poincaré's inequality

For a bounded Lipschitz domain, pointwise restriction to the boundary extends to a bounded linear trace operator

$$\mathrm{Tr} : H^1(\Omega) \longrightarrow H^{1/2}(\partial\Omega). \tag{9.3}$$

The space with homogeneous Dirichlet trace is

$$H^1_0(\Omega) = \overline{C_c^\infty(\Omega)}^{H^1(\Omega)} = \ker(\mathrm{Tr}).$$

Theorem 9.6 (Poincaré inequality)**.** *If Ω is bounded, connected, and Lipschitz, then there is a constant $C_P > 0$, depending only on Ω, such that*

$$\|v\|_{L^2(\Omega)} \le C_P \|\nabla v\|_{L^2(\Omega)} \qquad \forall v \in H^1_0(\Omega). \tag{9.4}$$

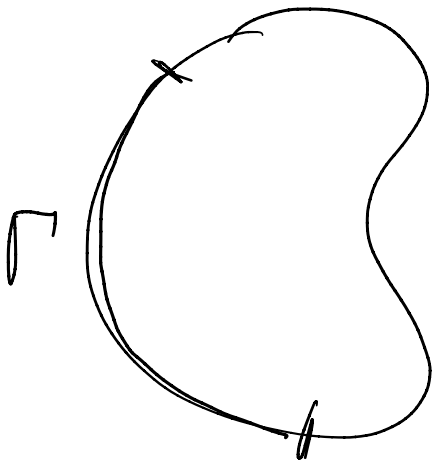
Consequently,

$$(u, v)_V = \int_\Omega \nabla u \cdot \nabla v \, dx, \qquad \|v\|_V = \|\nabla v\|_{L^2(\Omega)},$$

define an inner product and an equivalent norm on $V = H^1_0(\Omega)$.

$$H^1_\Gamma(\Omega) = \{ u \in H^1(\Omega) : u|_\Gamma = 0 \}$$

where $\Gamma \subset \partial\Omega$



- If $u \in H^1(\Omega)$, then from distributional derivative def + being in L^2

$$\int_\Omega u' \varphi \, dx = -\int_\Omega u \varphi' \, dx \qquad \forall \varphi \in C_c^\infty$$

If u has classical derivative, then it coincides with distributional (weak) derivative.

(Q ch. 1) General elliptic problems

General diff. operator in $\Omega \in \mathbb{R}^d$

$$\mathcal{L}u := -\nabla \cdot (A \nabla u) + \underline{b} \cdot \nabla u + c u$$

where

$$A \in L^\infty(\Omega; \mathbb{R}^{d\times d}), \quad \underline{b} \in L^\infty(\Omega; \mathbb{R}^d), \quad c \in L^\infty(\Omega)$$

$$A = (a_{ij})_{i,j=1}^d \qquad \underline{b} = (b_i)_{i=1}^d$$

test and integrate

$$a(u,v) = \int_\Omega \underbrace{\sum_{ij} a_{ij} D_j u D_i v}_{A \nabla u \cdot \nabla v} + \int_\Omega \underline{b} \cdot \nabla u \, v + \int_\Omega c u v$$

bilinear form

for $u, v \in H^1(\Omega)$

generalises Laplacian case $\mathcal{L}u = -\Delta u$ in which case $a(u,v) = \int_\Omega \nabla u \cdot \nabla v$.

$\rightarrow$ a is continuous in H^1

$$a(u,v) \le C \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}$$

Def: $\mathcal{L}$ is uniformly <u>elliptic</u> if

$A = \left(a_{ij}(x)\right)_{ij}$ is positive definite

almost everywhere in Ω. That is

$$(A\xi,\xi) = \sum_{ij} a_{ij}\xi_i\xi_j \ge \lambda|\xi|^2 \quad \forall \xi \in \mathbb{R}^d$$

$$= \xi^T A \xi$$

for $\exists \lambda > 0$.

Example: for $d=2$, and

$$\mathcal{L}u = a u_{xx} + c u_{yy} + 2b u_{xy} + \cdots$$

then

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

A is pos. def $\Leftrightarrow \begin{cases} a > 0 \\ ac - b^2 > 0 \end{cases}$

(or in ellipse $ax^2 + 2bxy + cy^2$)

Weak formulation

For the homogeneous Dirichlet problem:

$$\begin{cases} \mathcal{L}u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

The weak form reads:

find $u \in V := H_0^1(\Omega)$:

$$a(u,v) = F(v) \quad \forall v \in V$$

where $F(v) = \langle f, v \rangle$

for any $f \in V'$

and, in particular, if $f \in L^2(\Omega)$

$$= \int_\Omega f v \, dx$$

Lemma (Lax-Milgram): Let V a (real) Hilbert space w.r.t. $(\cdot,\cdot)_V$ (and $\|\cdot\|_V$ the induced norm); also

- $a(\cdot,\cdot) : V \times V \to \mathbb{R}$ bilinear form
  - * continuous:

$\exists \gamma > 0 : |a(w,v)| \le \gamma \|w\|_V \|v\|_V$
$\forall v, w \in V$

* coercive:

$\exists \alpha_0 > 0 : a(v,v) \ge \alpha_0 \|v\|_V^2$
$\forall v \in V$

- $F : V \to \mathbb{R}$ linear functional

* continuous: $\exists C_0 > 0$:

$|F(v)| \le C_0 \|v\|_V$
$\forall v \in V$

Then $\exists! \, u \in V$: $a(u,v) = F(v)$ (WP) $\forall v \in V$
(weak sol)

and $\|u\|_V \le \frac{1}{\alpha_0} \|F\|_{V'}$

That is, the weak problem (WP) is well posed.

Well-posedness for elliptic operators.

Assume $\mathcal{L}$ is elliptic; $V = H_0^1$

$$a(v,v) = \int_\Omega A\nabla v \cdot \nabla v + \underbrace{\int_\Omega \underline{b}\cdot\nabla v\, v}_{(II)} + \int_\Omega c v^2 \quad (III)$$

elliptic $\Rightarrow$

$$\alpha \|\nabla v\|^2_{L^2(\Omega)}$$

$$\int_\Omega \underline{b}\cdot\nabla v\, v = \frac{1}{2}\int_\Omega \underline{b}\cdot\nabla v^2 = -\int_\Omega \frac{1}{2}(\nabla\cdot\underline{b})\, v^2$$

$$(II)+(III) = \int_\Omega \left(c - \frac{1}{2}\nabla\cdot b\right) v^2$$

suff. condition

Assume $\boxed{c - \frac{1}{2}\nabla\cdot b \geq 0}$, then

$$a(v,v) \overset{①}{\geq} \alpha \|\nabla v\|^2_{L^2(\Omega)} = \alpha |v|^2_{H^1(\Omega)}$$

$$\|v\|^2_{L^2} \leq C_P^2 \|\nabla v\|^2_{L^2} \quad \text{Poincaré} \quad \Rightarrow \quad \|\nabla v\|^2_{L^2} \overset{②}{\geq} \frac{1}{C_P^2} \|v\|^2_{L^2}$$

using ② into ①

$$a(v,v) \geq \frac{\alpha}{C_P^2} \|v\|^2_{L^2} \quad ③$$

Adding ① and ③

$$\left(\frac{1}{\alpha} + \frac{C_P^2}{\alpha}\right) a(v,v) \geq \|\nabla v\|^2_{L^2} + \|v\|^2_{L^2} = \|v\|^2_{H^1}$$

$$\Rightarrow \boxed{a(v,v) \geq \frac{\alpha}{1+C_P^2} \|v\|^2_{H^1}}$$

$\Rightarrow$ coercivity in H^1 for

- elliptic operators
- with $c - \frac{1}{2} \nabla \cdot \underline{b} \geq 0$